Лекция №4

1) Декодирование по максимуму правдоподобия

$x - y$

$\{p(y|x), x \in X, y \in Y\}$

$C = \{c_i, i = 1, \dots, M\} \subset X^n$

$\bar{y} = (y_1, y_2, \dots, y_n)$ ф-я правдопод

$p(\bar{y} | \bar{c})$ $i = 1, \dots, M$

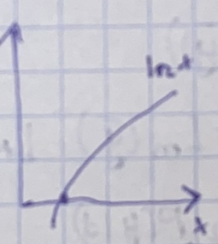
$p(\bar{y} | \bar{x}) = \prod_{i=1}^{n} p(y_i | x_i)$

логарифм ф.я

$\Lambda_m(\bar{y}) = \ln p(\bar{y} | \bar{c}_m) = \sum_{i=1}^{n} \ln p(y_i | c_{m_i}) = \sum_{i=1}^{n} \lambda_m(y_i)$

отношения правдоп.

$L(y) = \ln \frac{p(y|1) > 0}{p(y|0) < 0}$ $y = 1$, $y = 0$



$L(\bar{y}) = (L(y_1), L(y_2), \dots, L(y_n))$

$\bar{\varepsilon}_m = (\varepsilon_{m_1}, \varepsilon_{m_2}, \dots, \varepsilon_{m_n})$ $\varepsilon_{m_i} = 2 \cdot c_{m_i} - 1$

$c_{m_i} = 0 \to -1$

$c_{m_i} = 1 \to 1$

2) Т. Для ДСК декодир. по максимуму правдопод сводится к поиску кодового слова, для кот максимально:

$L_m(\bar{y}) = (\bar{\varepsilon}_m, L(\bar{y}))$

Док-во:

$\bar{C}_m$ и $C_{m'}$, $m \neq m'$ $\Lambda_m(\bar{y}) \geq \Lambda_{m'}(\bar{y}) \Rightarrow L_m(\bar{y}) \geq L_{m'}(\bar{y})$

$\downarrow$ $\swarrow$

$$\begin{array}{lll} & 1\ 2 & n \\ C_m = & (0\ 1 & \ldots\ 1\ 0) \\ C_{m'} = & (0\ 0 & \ldots\ 1\ 1) \end{array}$$

$I_{00} = \begin{matrix} C_m = 0 \\ C_{m'} = 0 \end{matrix}$

$I_{10} = \left\{ i \;\middle|\; \begin{matrix} C_{m_i} = 1 \\ C_{m'_i} = 0 \end{matrix} \right\}$ $\quad I_{11}, I_{01}$

$$L_m(\bar{y}) - L_{m'}(\bar{y}) = \sum_{j \in I_{10}} ((+1) - (-1)) \cdot L(y_j) + \sum_{j \in I_{01}} ((-1) - (+1)) \cdot L(y_j) =$$

$$= 2 \cdot \sum_{j \in I_{10}} L(y_j) - 2 \sum_{j \in I_{01}} L(y_j) \ominus$$

$$L(y_j) = \ln \frac{P(y_j|1)}{P(y_j|0)};$$

$$\ominus\ 2 \cdot \sum_{j = I_{10}} \ln \frac{P(y_j|\bar{C}_{m_j})}{P(y_j|C_{m_j})} + 2 \sum_{j \in I_{01}} \ln \frac{P(y_j|\bar{C}_{m'})}{P(y_j|\bar{C}_m)} =$$

$$= 2 \sum_{j=1}^{n} \ln \frac{P(y|C_{m_j})}{P(y|C_{m'_j})} = 2 \cdot \sum_{j=1}^{n} \ln P(y|C_{m_j}) - 2 \cdot \sum_{j=1}^{n} \ln P(y|C_{m'}) =$$

$$= 2 \cdot (\Lambda_m(\bar{y}) - \Lambda_{m'}(\bar{y}))$$

$$\Lambda_m(\bar{y}) - \Lambda_{m'}(y) \sim L_m(\bar{y}) - L_{m'}(\bar{y})$$

ДСК p_0 $\quad L(y) = \ln \frac{P(y|1)}{P(y|0)}$

$$L(y) = \begin{cases} \ln \frac{1 - p_0}{p_0}, & y = 1 \\ \ln \frac{p_0}{1 - p_0}, & y = 0 \end{cases}$$

$$L_m(\bar{y}) = (\xi_m, L(\bar{y})) \ominus$$

$y_i = C_{m_i} \to \delta$; $y_i = 1$ $C_{m_i} = 1 \to 1$; $y_i = 0$ $C_{m_i} = 0 \to -1$

$$\ominus \sum_{i=1}^{n} (2 \cdot C_{m_i} - 1) \cdot L(y_i).$$

$d(\bar{y}, \bar{C}_m)\cdot(-\delta) + (n - d(\bar{y}, \bar{C}_m))\cdot\delta = \delta(-d(\bar{y}, \bar{C}_m) + n - d(\bar{y}, \bar{C}_m)) =$

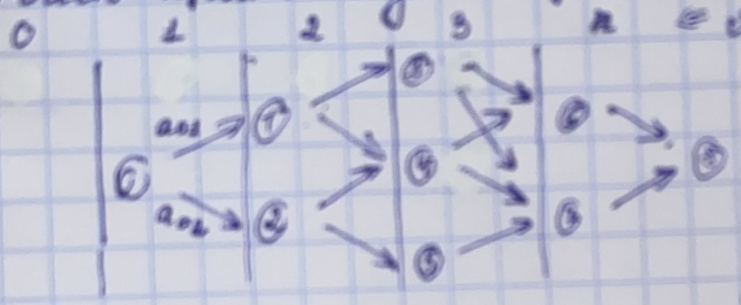
$= \delta\cdot(n - 2d(\bar{y}, \bar{C}_m)) \to max$

$\downarrow$

$p_0 < \frac{1}{2}, \delta > 0$ $\qquad$ min $\Rightarrow$ дек по min р. Хэмминга

$\Rightarrow$ по max плав. в ДОС

3) Поиск кратч. пути в реш. ①



Св-ва решетки:

- вершина разбита на непересек. мн-ва (ярусы)
- нулев. и последн. ярус содерж. ровно 1 узел
- решетка - направл. граф
- длина пути = Σ дл. ребер